

Operations Research & Applications programming

Ch. 3

The Simplex Algorithm

Introduction

- *In this lecture* we discuss the computational aspects of the Simplex algorithm.
- We shall see how a LPP is put into a simplex tableau. Starting from a BFS, we explain how to proceed step by step till we reach the optimal solution.
- The Mathematics behind the method (algorithm) will be seen only later. We illustrate the Simplex method by examples.

Computational Details of the Simplex Algorithm

□ *Example* : Consider the LPP

$$\text{Maximize } z = 8x_1 + 9x_2$$

Subject to $2x_1 + 3x_2 \leq 50$

$$2x_1 + 6x_2 \leq 80$$

$$3x_1 + 3x_2 \leq 70$$

$$x_1, x_2 \geq 0$$

□ Introducing *slack* variables, the LPP is same as

Computational Details of the Simplex Algorithm

$$\text{Maximize } z = 8x_1 + 9x_2 + 0s_1 + 0s_2 + 0s_3$$

Subject to the constraint :

$$2x_1 + 3x_2 + s_1 = 50$$

$$2x_1 + 6x_2 + s_2 = 80$$

$$3x_1 + 3x_2 + s_3 = 70$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

s_1, s_2, s_3 are *slack* variables

(*)

Computational Details of the Simplex Algorithm

- ❑ We see that if we take x_1, x_2 as *nonbasic variables*, a **BFS** can be got immediately as $(0,0, 50,80,70)$.
- ❑ Remember the first two components refer to x_1, x_2 and the last three components refer to s_1, s_2, s_3 .
- ❑ This is possible because in the LHS of the constraint equations (*) the coefficients of the basic variables s_1, s_2, s_3 form an Identity matrix (of size 3×3).

Computational Details of the Simplex Algorithm

- This is the first thing to be had to start the simplex method: the coefficients of the starting basic variables in the constraint equations form an **identity matrix** so that we can read out the starting **BFS**.
- Next we put the objective function and the constraint equations in a table (usually referred to as a Simplex tableau). For this we write the objective function also as an equation:

$$z - 8x_1 - 9x_2 + 0s_1 + 0s_2 + 0s_3 = 0$$

Computational Details of the Simplex Algorithm

- Then we build a Simplex tableau (Starting tableau)

Basic	z	x_1	x_2	s_1	s_2	s_3	Solution
z	1	-8	-9	0	0	0	0
s_1	0	2	3	1	0	0	50
s_2	0	2	6	0	1	0	80
s_3	0	3	3	0	0	1	70

Simplex tableau (Starting tableau)

- ❑ We note that the z-row is the *objective function row*.
- ❑ The remaining 3 rows are the *basic variable rows*. Each row corresponds to a basic variable; the left most variable denotes the basic variable corresponding to that row. Except for the legends (the topmost row and the leftmost column), all the other 4 rows are the objective function row and the constraint equations. Notice that in the objective function row, the coefficients of the basic variables are zero.

Simplex tableau (Starting tableau)

- This is the second requirement to start the simplex algorithm. Also we note that in each column corresponding to a basic variable, only that basic variable has a non-zero coefficient, namely 1, and all the other basic variables have zero coefficients.
- We see at present, the objective function, z , has value zero. We now seek to make one of the nonbasic variables as basic (and so) one of the basic

Simplex tableau (Starting tableau)

- variables will become nonbasic (that is will drop down to zero) and the nonbasic variable that will become basic is chosen such that the objective function will “improve”: i.e. in a maximization problem it will *increase* and in a minimization problem it will *decrease*. The nonbasic variable that will become basic is referred to as “*entering*” variable and the basic variable that will become nonbasic is referred to as “*leaving*” variable.

Simplex tableau (Starting tableau)

- ❑ Criterion for “*entering*” variable: (**Optimality Condition**).
- ❑ Choose that variable as the “*entering*” variable which has the **most –ve** coefficient in the z-row in case it is a **maximization** problem (as the variable which has the **most +ve** coefficient in the z-row in case it is a **minimization** problem).

(Break the ties arbitrarily.)

Simplex tableau (Starting tableau)

- ❑ Criterion for “leaving” variable: (**Feasibility Condition**).
- ❑ Let b_i be the RHS of the i^{th} row. Let a_{ij} be the coefficient of the entering variable x_j in the i^{th} row. The following “minimum ratio test” decides the leaving variable:
 - ❑ $\left\{ \frac{b_i}{a_{ij}} : a_{ij} > 0 \right\}$ is lest.

(Break the ties arbitrarily.)

Simplex tableau (Starting tableau)

- ❑ The entering variable column is called the **pivot column**. The leaving variable row is called the **pivot row**. The coefficient in the intersection of the two is referred to as the **pivot element**. Now we apply *elementary row operations* to modify the simplex tableau so that the pivot column has **1** at the pivot element and **zero** in all other places.
- ❑ We say *one iteration is over* and the new basic feasible solution and the new objective function value can be read from the new tableau. We illustrate by an example.

Simplex tableau (Starting tableau)

Enters

Leaves

Basic	z	x_1	x_2	s_1	s_2	s_3	Solution
z	1	-8	-9	0	0	0	0
s_1	0	2	3	1	0	0	50
s_2	0	2	6	0	1	0	80
s_3	0	3	3	0	0	1	70

Pivot element

Simplex tableau (Starting tableau)

- The elementary Row operations needed to produce the new basic solution include two types.

1. Pivot row

- Replace the leaving variable in the *Basic* column with the entering variable.
- New pivot row = Current pivot row / Pivot element

2. All other rows, including z

New Row = (Current row) - (Its pivot column coefficient) x (New pivot row)

- These computations are applied to the preceding tableau in the following manner :

Simplex tableau (Starting tableau)

- 1. Replace s_1 in the *Basic* column with x_1 :

$$\text{New } x_2\text{-row} = \text{Current } s_2\text{-row} / 6 = (0 \ 1/3 \ 1 \ 0 \ 1/6 \ 0 \ 40/3)$$

- 2. New $z\text{-row} = \text{Current } z\text{-row} - (-9) \times \text{New } x_2\text{-row}$
 $= (1 \ -8 \ -9 \ 0 \ 0 \ 0 \ 0) - (-9) \times (0 \ 1/3 \ 1 \ 0 \ 1/6 \ 0 \ 40/3)$
 $= (1 \ -5 \ 0 \ 0 \ 3/2 \ 0 \ 120)$

- 3. New $s_1\text{-row} = \text{Current } s_1\text{-row} - (3) \times \text{New } x_2\text{-row}$
 $= (0 \ 2 \ 3 \ 1 \ 0 \ 0 \ 50) - (3) \times (0 \ 1/3 \ 1 \ 0 \ 1/6 \ 0 \ 40/3)$
 $= (0 \ 1 \ 0 \ 1 \ -1/2 \ 0 \ 10)$

- 4. New $s_3\text{-row} = \text{Current } s_3\text{-row} - (3) \times \text{New } x_2\text{-row}$
 $= (0 \ 3 \ 3 \ 0 \ 0 \ 1 \ 70) - (3) \times (0 \ 1/3 \ 1 \ 0 \ 1/6 \ 0 \ 40/3)$
 $= (0 \ 2 \ 0 \ 0 \ -1/2 \ 1 \ 30)$

Simplex tableau (Starting tableau)

- Performing the elementary Row operations, we get the new Simplex tableau below

		Enters						
Basic		z	x ₁	x ₂	s ₁	s ₂	s ₃	Solution
z		1	-5	0	0	3/2	0	120
Leaves ← s ₁		0	1	0	1	-1/2	0	10
x ₂		0	1/3	1	0	1/6	0	40/3
s ₃		0	2	0	0	-1/2	1	30

Pivot element

Simplex tableau

- Performing the elementary Row operations, we get the new Simplex tableau below

Basic	z	x_1	x_2	s_1	s_2	s_3	Solution
z	1	0	0	5	-1	0	170
x_1	0	1	0	1	-1/2	0	10
x_2	0	0	1	-1/3	1/3	0	10
s_3	0	0	0	-2	1/2	1	10

Enters



Leaves



Pivot element

Simplex tableau

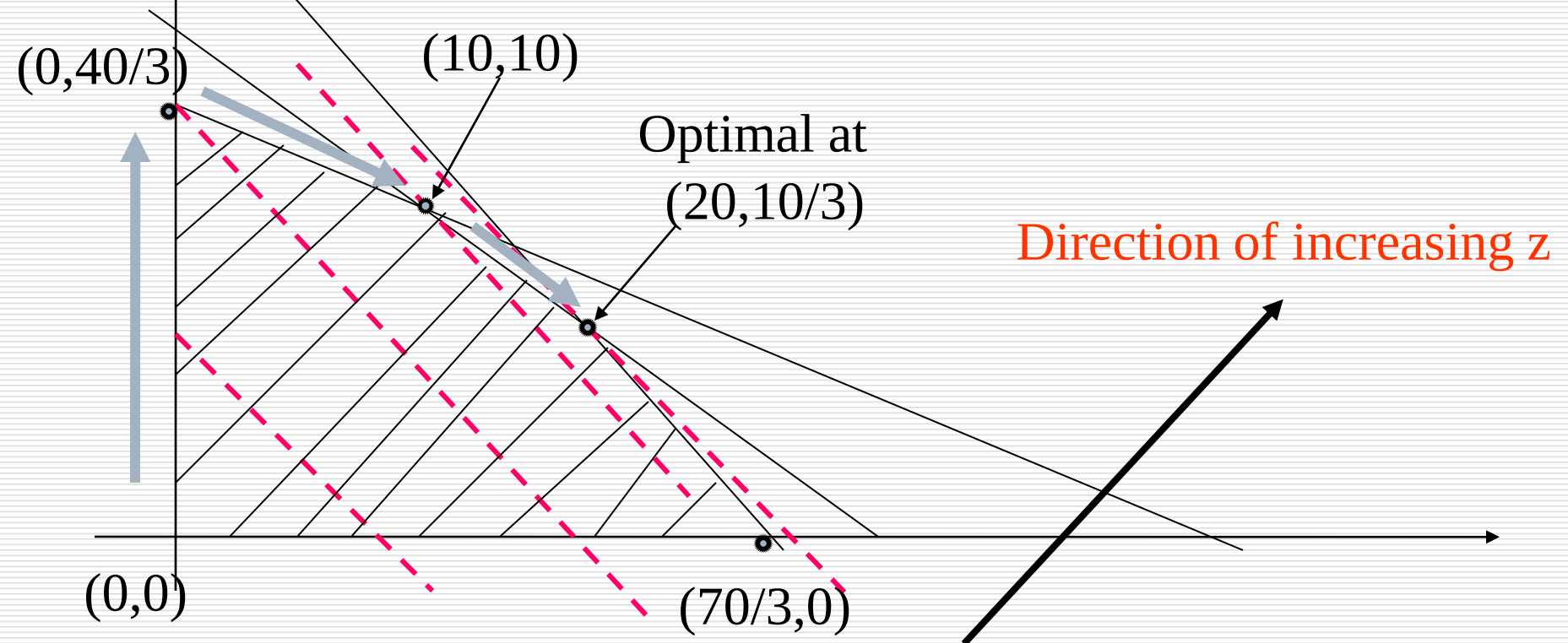
- Performing the elementary Row operations, we get the new Simplex tableau below:

Basic	z	x_1	x_2	s_1	s_2	s_3	Solution
z	1	0	0	1	0	2	190
x_1	0	1	0	-1	0	1	20
x_2	0	0	1	1	0	-2/3	10/3
s_2	0	0	0	-4	1	2	20

- This is optimal as all entries in z row are ≥ 0 .
- Optimum value = 190 at $x_1=20$, $x_2=10/3$

Graphical Solution of the problem

□ Consider the LPP



Summary of the Simplex Method

- *Optimality condition.* The **entering** variable in a maximization (minimization) problem is the *nonbasic* variable having the *most* negative (positive) coefficient in the z-row. Ties are broken arbitrarily. The optimum is reached at the iteration where all the z-row coefficients of the nonbasic variables are nonnegative (nonpositive).
- *Feasibility condition.* For both the maximization and the minimization problems, the **leaving variable** is the *basic variable* associated with the *smallest nonnegative ratio* (**with strictly positive denominator**). Ties are broken arbitrarily.

Summary of the Simplex Method

□ The steps of the simplex method are:

Step 1. Determine a starting basic feasible solution.

Step 2. Select an *entering variable* using the optimality condition. Stop if there is no entering variable; the last solution is optimal. Else, go to *step3*.

Step 3. Select a *leaving variable* using the feasibility condition.

Step 4. Determine the new basic solution by using the appropriate Gauss-Jordan computations. Go to *step2*.

Example :

□ Consider the following LPP :

$$\text{Maximize } z = x_1 + 2x_2 + 4x_3$$

Subject to the constraints

$$3x_1 + x_2 + 5x_3 \leq 10$$

$$x_1 + 4x_2 + x_3 \leq 8$$

$$2x_1 + 2x_3 \leq 7$$

$$x_1, x_2, x_3 \geq 0$$

Example :

□ Introducing slack variables, the LPP becomes :

$$\text{Maximize } z = x_1 + 2x_2 + 4x_3 + 0s_1 + 0s_2 + 0s_3$$

Subject to the constraints

$$3x_1 + x_2 + 5x_3 + s_1 = 10$$

$$x_1 + 4x_2 + x_3 + s_2 = 8$$

$$2x_1 + 2x_3 + s_3 = 7$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

Example :

Basic	z	x1	x2	x3 ↓	s1	s2	s3	Sol.
z	1	-1	-2	-4	0	0	0	0
s1	0	3	1	5	1	0	0	10
s2	0	1	4	1	0	1	0	8
s3	0	2	0	2	0	0	1	7
z	1	7/5	-6/5 ↓	0	4/5	0	0	8
x ₃	0	3/5	1/5	1	1/5	0	0	2
s2	0	2/5	19/5	0	-1/5	1	0	6
s3	0	4/5	-2/5	0	-2/5	0	1	3
z	1	29/19	0	0	14/19	6/19	0	188/19
x3	0	11/19	0	1	4/19	-1/19	0	32/19
x2	0	2/19	1	0	-1/19	5/19	0	30/19
s3	0	16/19	0	0	-8/19	2/19	1	69/19

Example :

□ The last tableau is the optimal tableau as all entries in the objective function row are ≥ 0 and the LPP is a maximization problem.

□ Optimal Solution is

$$x_1 = 0, x_2 = 30/19, x_3 = 32/19$$

And the Optimal z (= Maximum z) = $188/19$

Problem 2(a) Problem set 3.3B Page 100

□ Consider the following LPP :

$$\text{Maximize } z = 2x_1 + x_2 - 3x_3 + 5x_4$$

Subject to the constraints

$$x_1 + 2x_2 - 2x_3 + 4x_4 \leq 40$$

$$2x_1 - x_2 + x_3 + 2x_4 \leq 8$$

$$4x_1 - 2x_2 + x_3 - x_4 \leq 10$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Problem 2(a) Problem set 3.3B Page 100

□ Introducing slack variables, the LPP becomes :

$$\text{Maximize } z = 2x_1 + x_2 - 3x_3 + 5x_4 + 0s_1 + 0s_2 + 0s_3$$

Subject to the constraints

$$x_1 + 2x_2 - 2x_3 + 4x_4 + s_1 = 40$$

$$2x_1 - x_2 + x_3 + 2x_4 + s_2 = 8$$

$$4x_1 - 2x_2 + x_3 - x_4 + s_3 = 10$$

$$x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0$$

Problem 2(a) Problem set 3.3B Page 100

Basic	z	x1	x2	x3	x4	s1	s2	s3	Sol.
z	1	-2	-1	3	-5	0	0	0	0
s1	0	1	2	-2	4	1	0	0	40
s2	0	2	-1	1	2	0	1	0	8
s3	0	4	-2	1	-1	0	0	1	10
z	1	3	-7/2	11/2	0	0	5/2	0	20
s1	0	-3	4	-4	0	1	-2	0	24
x4	0	1	-1/2	1/2	1	0	1/2	0	4
s3	0	5	-5/2	3/2	0	0	1/2	1	14
z	1	3/8	0	2	0	7/8	3/4	0	41
x2	0	-3/4	1	-1	0	1/4	-1/2	0	6
x4	0	5/8	0	0	1	1/8	1/4	0	7
s3	0	25/8	0	-1	0	5/8	-3/4	1	29

Problem 2(a) Problem set 3.3B Page 100

□ The last tableau is the optimal tableau as all entries in the objective function row are ≥ 0 and the LPP is a maximization problem.

□ Optimal Solution is

$$x_1 = 0, x_2 = 6, x_3 = 0, x_4 = 7$$

And the Optimal z (= Maximum z) = 41

Special Cases in the Simplex Method

- The This section considers four special cases that arise in the use of the simplex method.
 1. Degeneracy.
 2. Alternative optima.
 3. Unbounded solutions.
 4. Nonexisting (or infeasible) solutions.
- Our interest in studying these special cases is twofold: (1) to present a *theoretical* explanation of these situations and (2) to provide a *practical* interpretation of what these special results could mean in a real-life problem.

Special Cases in the Simplex Method

□ The Tie for leaving variable - Degeneracy

- If in a Simplex iteration, there is a tie for leaving variable, then the resulting BFS will be degenerate, in the sense that one of the basic variables will also be zero. This may result in getting a new BFS without improving z . And in some cases this may lead to “cycling.”

Example 3.5-1 (Degenerate Optimal Solution)

- Consider the LPP

$$\text{Maximize } z = 3x_1 + 9x_2$$

Subject to $x_1 + 4x_2 \leq 8$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

- Introducing *slack* variables, the LPP is same as

Example 3.5-1 (Degenerate Optimal Solution)

$$\text{Maximize } z = 3x_1 + 9x_2 + 0s_1 + 0s_2$$

Subject to the constraint :

$$x_1 + 4x_2 + s_1 = 8$$

$$x_1 + 2x_2 + s_2 = 4$$

(*)

$$x_1, x_2, s_1, s_2 \geq 0$$

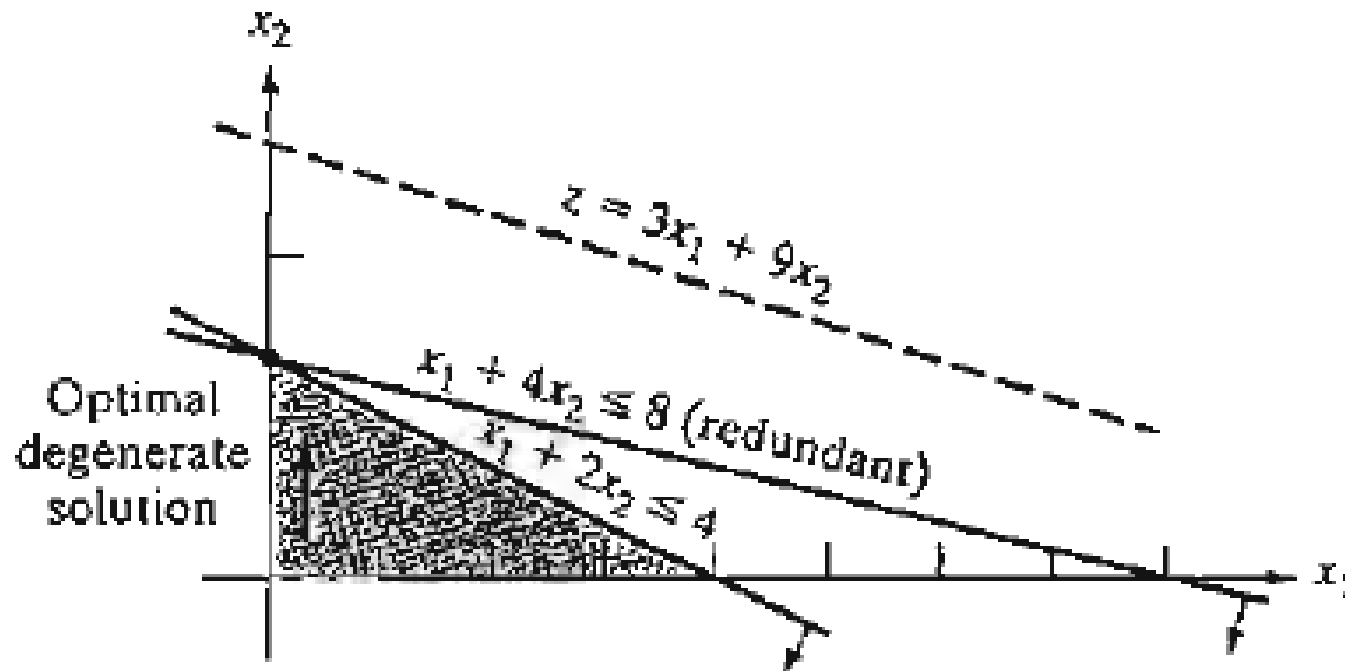
s_1, s_2 are *slack* variables

Example 3.5-1 (Degenerate Optimal Solution)

Basic	z	x1	x2	s1	s2	Sol.
z	1	-3	-9	0	0	0
s1	0	1	4	1	0	8
s2	0	1	2	0	1	4
z	1	-3/4	0	9/4	0	18
x2	0	1/4	1	1/4	0	2
s2	0	1/2	0	-1/2	1	0
z	1	0	0	3/2	3/2	18
x2	0	0	1	1/2	-1/2	6
x1	0	1	0	-1	2	0

Graphical Solution of the problem

□ LP degeneracy in Example 3.5-1



Special Cases in the Simplex Method

□ Multiple (Alternative) Optimal Solutions

- If Suppose in the optimal tableau, some non basic variable has zero coefficient in the objective function row, then that variable can become basic and we would get another BFS. Since the coefficient of the entering variable in the objective function row. The objective function does not improve. Hence the new BFS we get is an alternative Optimal Solution. Usually you are supposed to give all Optimal Solutions

Example 3.5-2 (Infinite number of Solution)

□ .Consider the LPP

$$\text{Maximize } z = 2x_1 + 4x_2$$

Subject to $x_1 + 2x_2 \leq 5$

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

□ Introducing *slack* variables, the LPP is same as

Example 3.5-2 (Infinite number of Solution)

$$\text{Maximize } z = 2x_1 + 4x_2 + 0s_1 + 0s_2$$

Subject to the constraint :

$$x_1 + 2x_2 + s_1 = 5$$

$$x_1 + x_2 + s_2 = 4$$

(*)

$$x_1, x_2, s_1, s_2 \geq 0$$

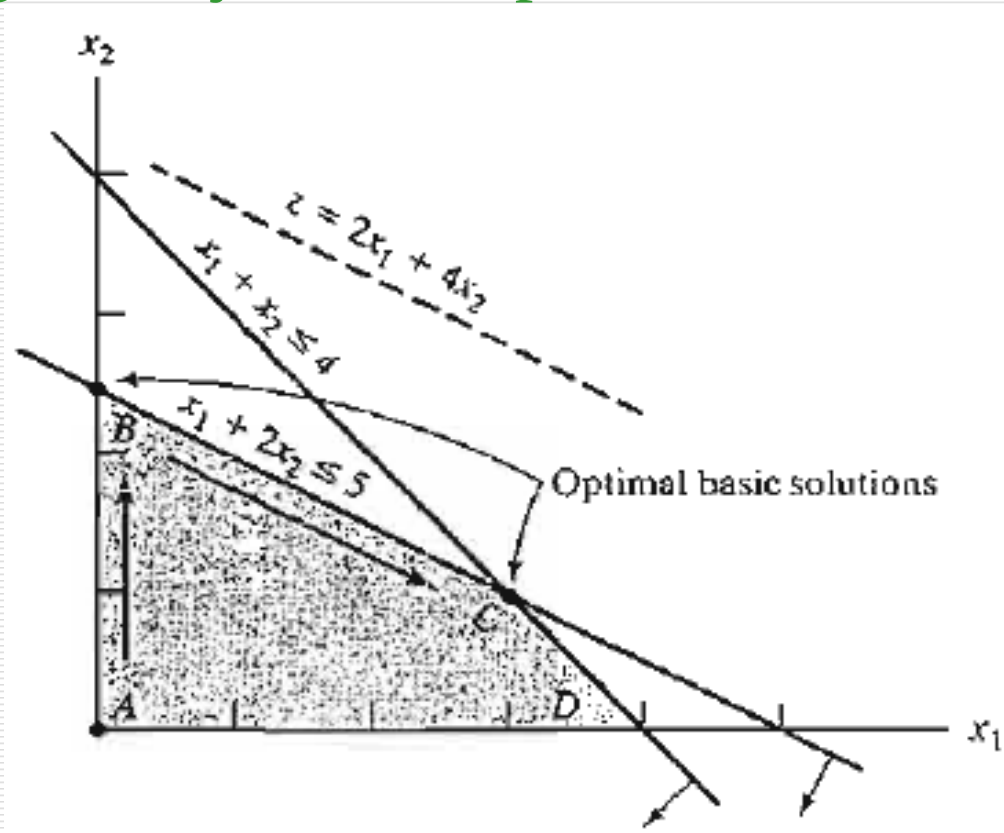
s_1, s_2 are *slack* variables

Example 3.5-2 (Infinite number of Solution)

Basic	z	x1	x2	s1	s2	Sol.
z	1	-2	-4	0	0	0
s1	0	1	2	1	0	5
s2	0	1	1	0	1	4
z	1	0	0	2	0	10
x2	0	1/2	1	1/2	0	5/2
s2	0	1/2	0	-1/2	1	3/2
z	1	0	0	2	0	10
x2	0	0	1	1	-1	1
x1	0	1	0	-1	2	3

Graphical Solution of the problem

□ LP degeneracy in Example 3.5-2



Special Cases in the Simplex Method

- No leaving variable -Unbounded Solution
- Suppose in a Simplex iteration, some variable can enter but no variable can leave. This will happen when all the entries in the entering variable column are either zero or negative, This means the entering variable (which is right now having the zero value) can be increased indefinitely without violating any of the constraints and thus z can be increased indefinitely or the problem has unbounded solution.

Example 3.5-3 (Unbounded objective value)

- Consider the LPP

$$\text{Maximize } z = 2x_1 + x_2$$

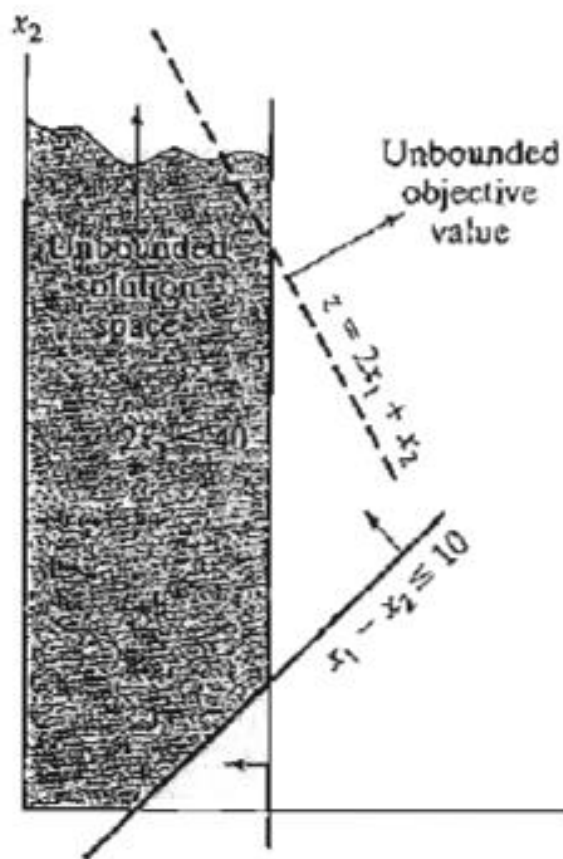
Subject to $x_1 - x_2 \leq 10$

$$2x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

- Introducing *slack* variables, the LPP is same as

Example 3.5-3 (Unbounded objective value)



Basic	z	x1	x2	s1	s2	Sol.
z	1	-2	-1	0	0	0
s1	0	1	-1	1	0	10
s2	0	2	0	0	1	40
z	1	0	-3	2	0	20
x1	0	1	-1	1	0	10
s2	0	0	2	-2	1	20
z	1	0	0	-1	3/2	50
x1	0	1	0	0	1/2	20
x2	0	0	1	-1	1/2	10

Special Cases in the Simplex Method

☐ Infeasible Solution

- ☐ Suppose LP models with inconsistent constraints have no feasible solution. This situation can never occur if *all* the constraints are of the type $\leq$ with nonnegative right-hand sides because the slacks provide a feasible solution.

Problem

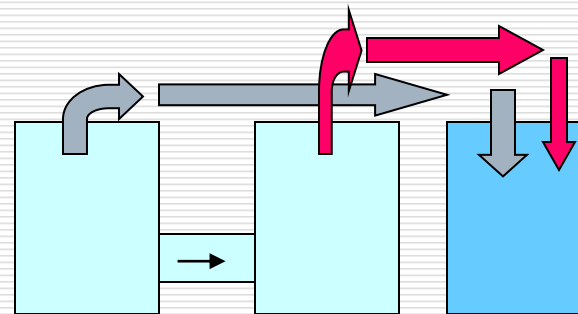
- ❑ Shale Oil, located on the island of Aruba, has a capacity of 600,000 barrels of crude oil per day. The final products from the refinery include two types of unleaded gasoline: regular and premium. The refining process encompasses three stages: (1) a distillation tower that produces feedstock, (2) a cracker unit that produces gasoline stock by using a portion of the feedstock produced from the distillation tower, and (3) a blender unit that blends the gasoline stock from the cracker unit and the feedstock from the distillation tower.

Problem

- ❑ Both regular and premium gasoline can be blended from either the feedstock or the gasoline stock at different production costs. The company estimates that the net profit per barrel of regular gasoline is \$7.70 and \$5.20, depending on whether it is blended from feedstock or from gasoline stock. The corresponding profit values for the premium grade are \$10.40 and \$12.30.
- ❑ Design specifications require 5 barrels of crude oil to produce 1 barrel of feedstock. The capacity of the

Problem

- cracker unit is 40,000 barrels of feedstock a day. All remaining feedstock is directly used in the blender unit to produce end-product gasoline. The demand limits for regular and premium gasoline are 80,000 and 50,000 barrels per day. Develop a model for determining the optimum production schedule for the refinery.



Distillation Tower Cracker Unit Blender Unit

Solution

- ❑ Let x_1 units from feedstock be used to blend (= make) regular gasoline.
- ❑ Let x_2 units from feedstock be used to blend (= make) premium gasoline.
- ❑ Let x_3 units from gasoline stock be used to blend (= make) regular gasoline.
- ❑ Let x_4 units from gasoline stock be used to blend (= make) premium gasoline.

Problem 2(a) Problem set 3.3B Page 100

□ Thus we have to maximize the profit :

$$z = 7.70x_1 + 10.40x_2 + 5.20x_3 + 12.30x_4$$

Subject to the constraints

$$x_1 + x_2 + x_3 + x_4 \leq 120,000$$

$$x_3 + x_4 \leq 40,000$$

$$x_1 + x_3 \leq 80,000$$

$$x_2 + x_4 \leq 50,000$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Problem

□ Optimum Solution

$$x_1 = 70,000$$

$$x_2 = 10,000$$

$$x_3 = 0$$

$$x_4 = 40,000$$

$$\text{Maximum } z = 1135000$$